INDICATION: A 61-year-old female sustained a displaced, intra-articular distal radius fracture of the left wrist following a fall from standing height onto an outstretched hand (Colles-type intra-articular fracture, AO type 23-C2).

PROCEDURE PERFORMED: Open reduction and internal fixation (ORIF) of the left distal radius using a Synthes 2.4 mm variable-angle locking volar distal radius plate.

SURGICAL APPROACH AND FIXATION: After induction of intravenous Bier block regional anesthesia and administration of 2 g cefazolin, the left upper extremity was prepped and draped in the usual sterile fashion. A pneumatic forearm tourniquet was inflated to 220 mmHg. A modified Henry approach was utilized. An 8 cm longitudinal incision was made along the radial border of the flexor carpi radialis (FCR) tendon. The FCR tendon sheath was opened, and the tendon retracted medially while radial artery was identified, mobilized, and protected laterally.

The underlying pronator quadratus muscle was sharply incised at its radial insertion leaving a 2 mm cuff for subsequent repair, and reflected subperiosteally to expose the volar cortex of the distal radius. The fracture fragments were cleaned of organizing hematoma and fibrous tissue. Under fluoroscopic guidance, longitudinal traction with wrist palmar-flexion and ulnar deviation achieved anatomical reduction of the dorsal tilt, radial height, and articular surface. K-wires were placed temporarily to secure the radial styloid and lunate facet fragments.

A pre-contoured 2.4 mm variable-angle volar locking plate was positioned proximal to the watershed line. A cortical screw was inserted through the oblong gliding hole in the shaft to establish height and plate alignment. Intraoperative fluoroscopy confirmed excellent restoration of radial inclination (22 degrees), volar tilt (11 degrees), and complete resolution of articular step-off. Multiple subchondral locking screws were systematically placed across the distal articular row, followed by proximal cortical locking screws into the diaphysis. Final anteroposterior, lateral, and 20-degree elevated traction views demonstrated anatomical articular reduction without screw penetration into the radiocarpal joint. The pronator quadratus was repaired over the plate with interrupted 3-0 Vicryl. The skin was closed with 4-0 Monocryl subcuticular suture and steristrips. A well-padded volar wrist splint in neutral position was applied.